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# Libraries vs Frameworks: Key difference

**A simple way to remember it: you call a library. A framework calls you.**

**A library: a toolbox to help you develop**

A development library refers to an inventory of classes or functions designed to save time so you do not have to reinvent the wheel. It functions just like a toolbox on a building site: rather than spending precious project time building your own hammer from scratch, you pick up a ready-made tool and get straight to work.

**Framework: a ready-made software infrastructure**

A framework is not a library, though it often includes one or more of them. It provides an overall software infrastructure or operating model (such as MVC or MVVM) composed of autonomous software components. While a library is a toolbox, a framework is a laboratory: it provides tools, raw materials (variables and parameters), and safety rules—the strict formalism and paradigm you must follow.

- **Scope and weight:** Frameworks vary in size. Comprehensive ones like Angular can be heavy (around 150 kB), slowing page load times in the front end, whereas targeted alternatives like Mithril ship only what is necessary (8 kB for SPAs).
- **Complexity:** More features mean a steeper learning curve, requiring effort to navigate before finding the tools you need.
- **Advantages:** Time savings, guided workflows, common architectural models, and community support.
- **Disadvantages:** Rigid rules with little freedom, rarely a 100% fit for every need, and a learning curve that can be challenging for junior developers.

**The core difference: who calls what?**

The defining difference lies in control:

- **With a framework, the framework calls your code.** It wraps around your work and dictates the structure (such as splitting logic into View, Model, and Controller). You are merely a user supplying the data; the framework manages the rest.
- **With a library, your code calls the library.** You remain in control, pulling tools from the inventory whenever you need them.

**Open source and licensing**

Most libraries and frameworks are free and open-source (such as React), enabling public use and contributions. However, some are paid proprietary tools (such as Kendo UI) that restrict redistribution. Checking the license before use is essential to avoid legal trouble.

**Reference:**


1. https://www.numendo.com/en/blog/what-is-the-difference-between-a-framework-and-a-library/
2. https://hackmamba.io/engineering/libraries-vs-frameworks-which-is-right-for-your-next-web-project/